

As well as libraries, many organizations use databases to store records of staff, clients, or stock. Numerous websites, particularly those where people can keep adding content, such as social media sites, also use databases. Shops require databases to enable customers to buy items online. As databases store a great deal of sensitive information, it's very important to ensure they have adequate levels of security and are properly backed up.



Hospitals

Social media

Government departments

Schools

Banks

△ Who uses databases?

TOP TECH

More than 600 million tweets are posted every day. By analysing them as they happen, Twitter is able to identify and highlight trends in what is being discussed. People are often alerted to an event when it first starts trending on Twitter. Similarly, the Google Trends tool gives users access to data on Google searches. It's possible to see graphs of how often people across the world searched for a particular topic, or top 10 lists of popular searches in different categories. Both tools give users a picture of how people in their own vicinity and other countries are reacting to events.

Massive and fast-changing collections of data, from sources as diverse as social media to scientific experiments like the Large Hadron Collider, are being created and added to every day. This data presents a challenge to traditional databases and methods of analysis. Visualization, where data is presented as graphs or images, is one aspect of big data that has become increasingly important. This can help more people to better understand patterns found in the data, which can help in everything from designing medicines to forming governmental policies.



The interconnected network of physical objects with embedded sensors and internet connectivity that is called the Internet of Things is another source of big data.